

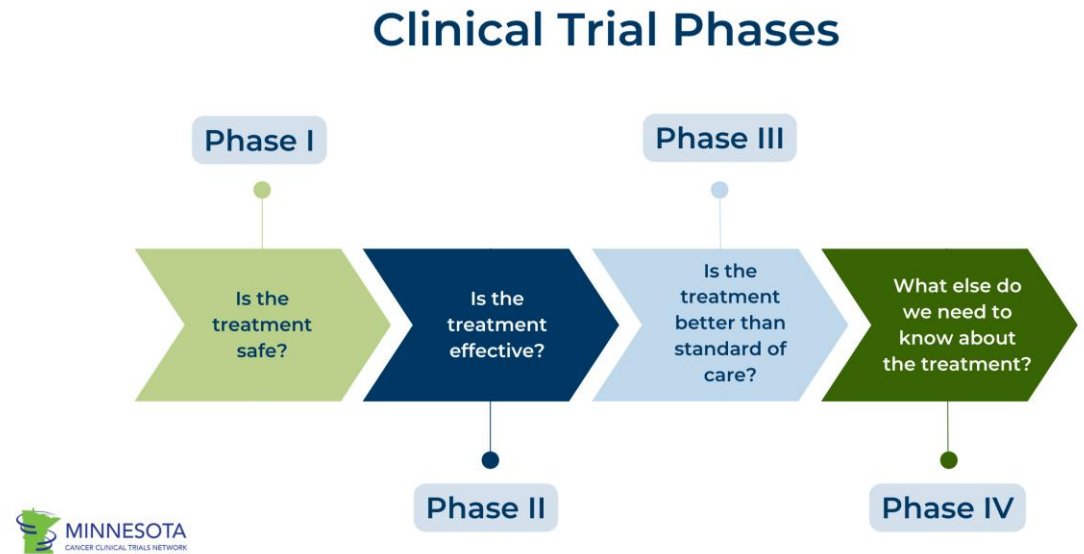
Stratified Randomization in Clinical Trials

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What is a Clinical Trial?

- A study that tests an experimental treatment to see if it is better than current treatment
 - Goal is to create a new standard of care
 - If study is effective, public can use treatment
- How to quantify?
 - Take difference in mean outcomes between treatment and control



Hypothesis Testing in Trials

Null hypothesis = H_0

- Difference in actual and expected means is zero

Alternative hypothesis = H_A

- Difference in means is greater or less than zero

In a clinical trial:

- H_0 : Current standard of care
- H_A : New treatment makes a difference

Sample Clinical Trial: Moderna COVID Vaccine



Goal: Create a vaccine that prevents symptomatic COVID



Randomly assigned placebo (saline) or vaccine



Measured and compared rates of positive COVID tests in both groups

Sample Clinical Trial: Moderna COVID Vaccine

H_0

- Efficacy of vaccine is 30% or less

H_A

- Efficacy of vaccine is above 30%

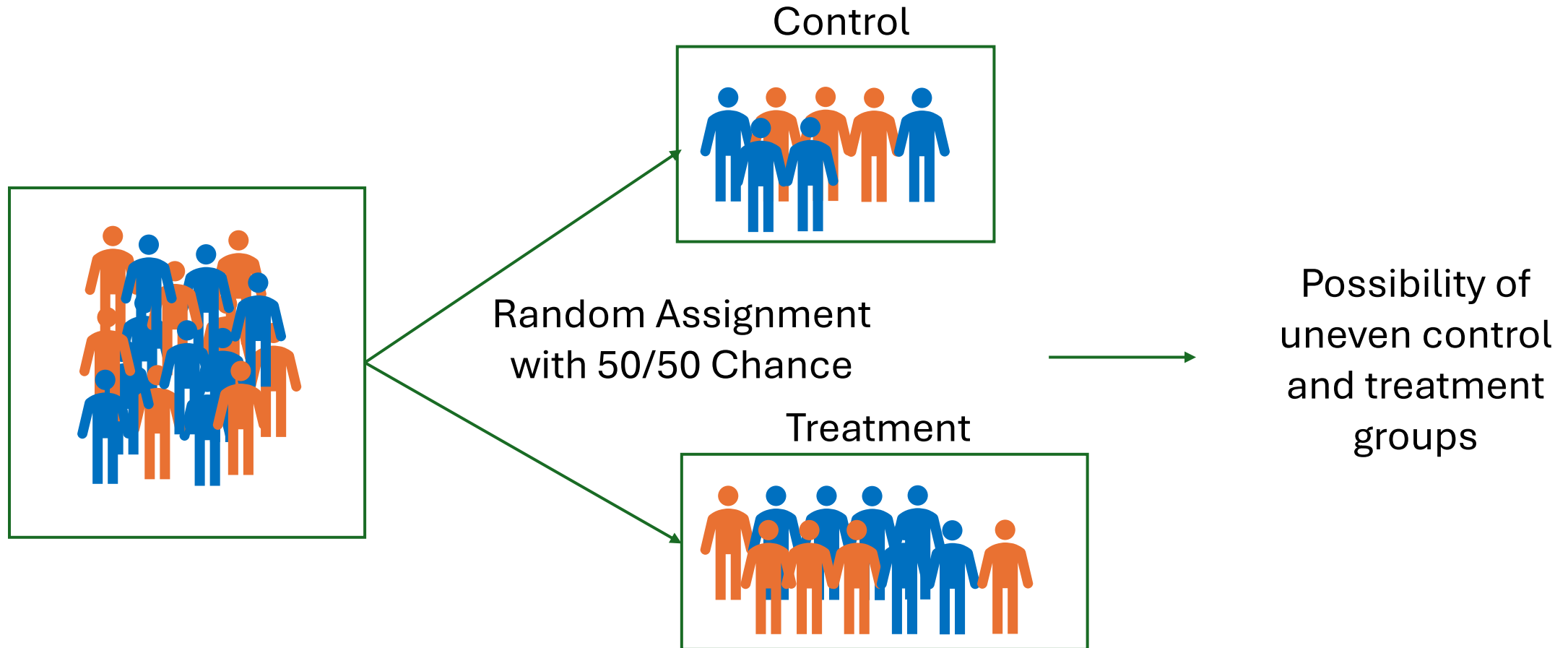
Tested at a significance level (α) of 0.001

- α = the chance of seeing results assuming the null hypothesis is true

Results:

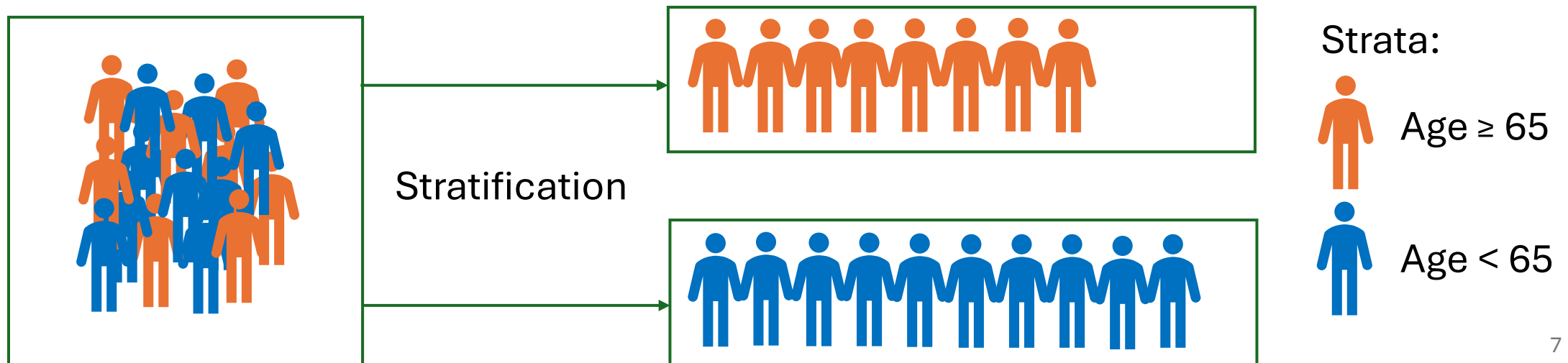
94.1% efficacy was found in preventing COVID infection

Random Assignment: Simple Randomization

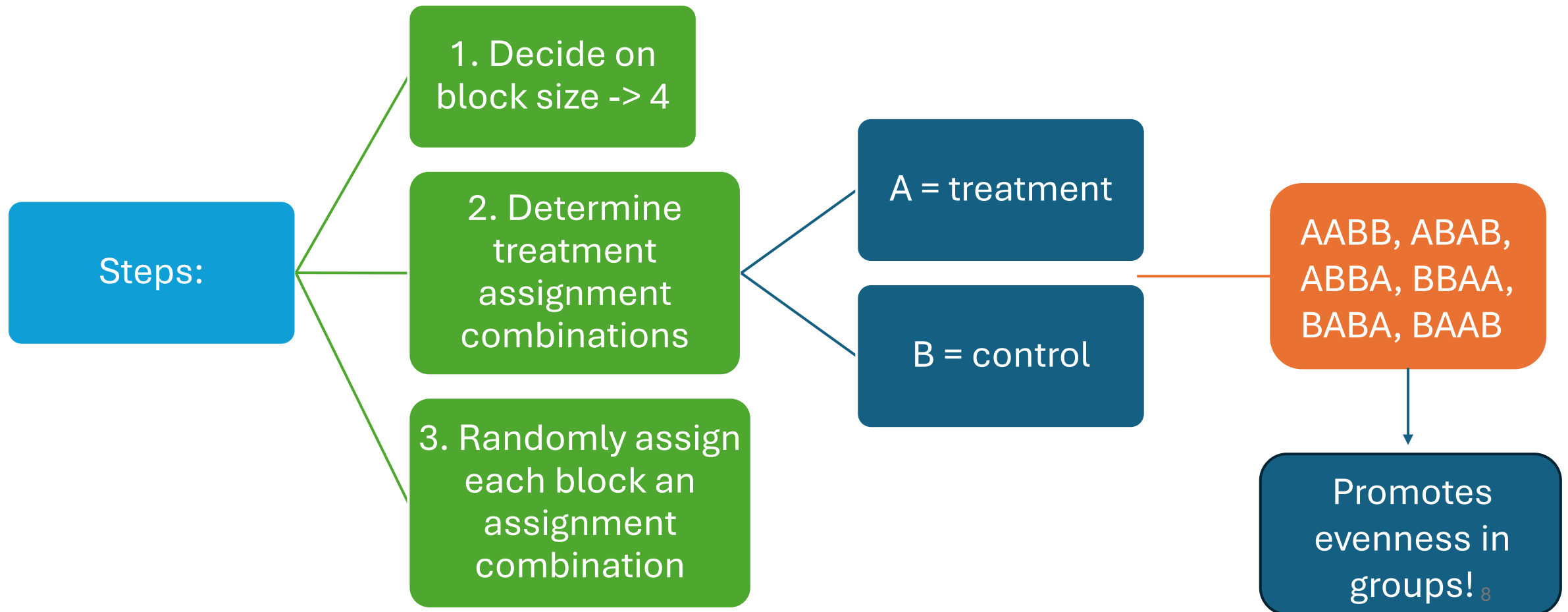


Stratification

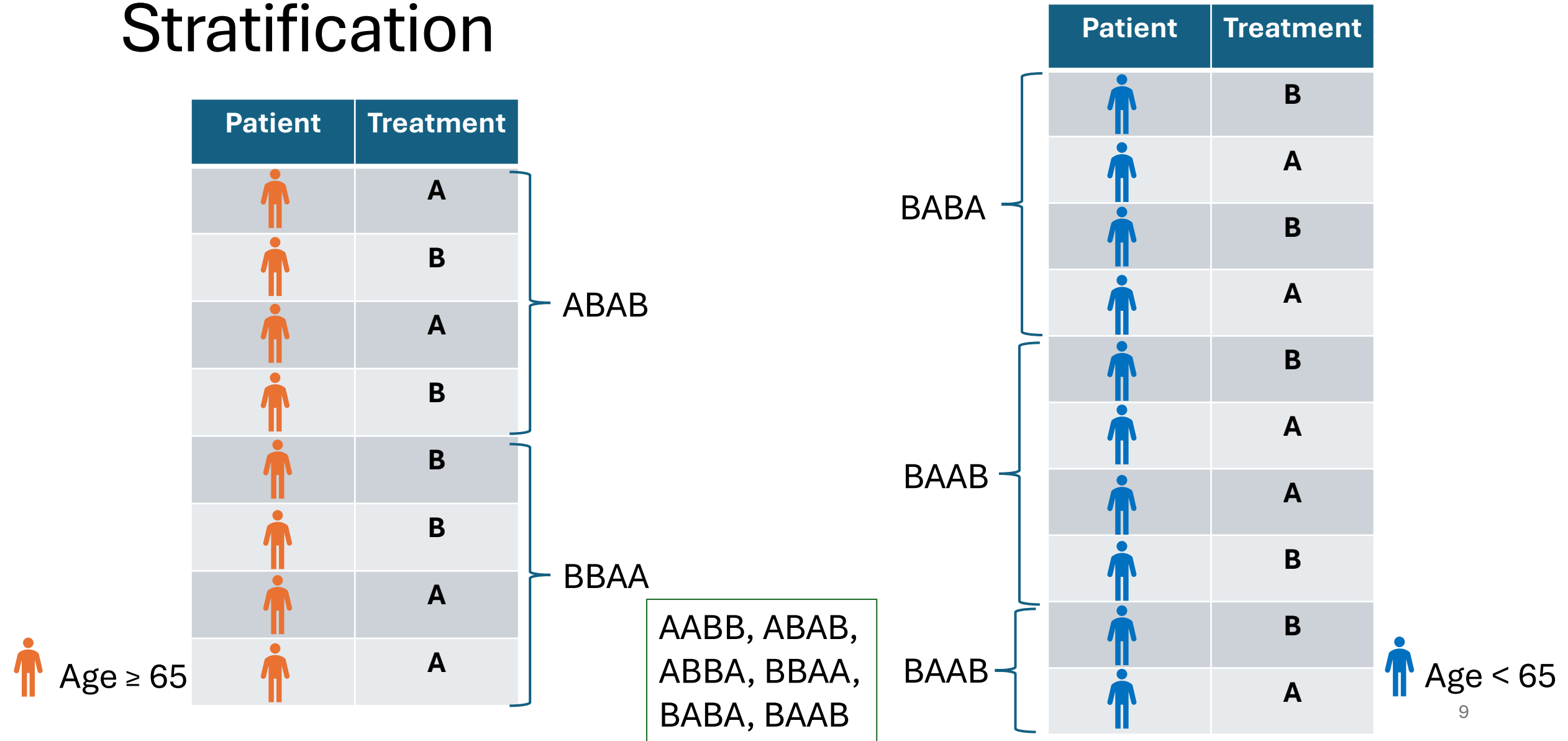
- Strata = a subset of the population based on a characteristic
 - Examples: age, gender, etc.
- Aims to avoid influence of covariates
 - Covariate = a variable that is not of interest, but may impact the response
- Goal = increase precision in clinical trials



Random Assignment: Stratified Block Randomization



Random Assignment: Permuted Block Stratification



What is Power and Why is it Good?

- Statistical power is related to sensitivity ->
 - The likelihood of a significance test detecting an effect where there is one
- Power is the probability of avoiding a type II error (rejecting the null hypothesis when you should not)
- Higher power -> larger chance of a test detecting a true effect of the treatment
- Influenced by
 - Sample size
 - Effect size
 - Significance level
- Stratification increasing power is proven by Model-Robust Inference for Clinical Trials that Improve Precision by Stratified Randomization and Covariate Adjustment by Wang, B., Susukida, R., Mojtabai, R., Amin-Esmaeili, M., & Rosenblum, M.

Conclusion

- Stratification in clinical trials divides the population into subgroups to:
 - Account for covariates and potential confounding variables
 - Reduce variance
 - Increase power
- Overall, aims to increase understanding of treatment efficacy